Jackfruit Phase-1

Software Requirements Specification

for

Smart Complaint Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

Introduction

Purpose

*The purpose of this SRS document is to define the functional and nonfunctional requirements of the project titled **Smart Complaint Management System**. The system is designed to provide a centralized platform through which users can register complaints, categorize and track their complaints, receive notifications regarding complaint progress, and view complaint status.*

The system also provides administrative capabilities for managing complaints, including complaint assignment, escalation, status updates, resolution, search and filtering which, dashboard monitoring, and reports and analytics. In addition, the system incorporates smart complaint classification to support priority categorization of the complaints for urgency to tackle and solve the problems and also manages duplicate complaint detection.

This document specifies the expected behavior, features, user interactions, operating environment, constraints, and quality requirements of the proposed system. It serves as a reference for the development, testing, validation, and maintenance of the Smart Complaint Management System.

The SRS covers Version 1.0 of the proposed system.

Intended Audience

This SRS is intended for the following stakeholders:

i)Developers: used to understand the functional requirements, system behavior, constraints, interfaces, and operating environment required for implementing the Smart Complaint Management System

ii)Project Manager:used to understand the overall scope, system features, priorities, dependencies, and requirements of the project for planning and monitoring development activities.

iii) Testers:they can use the functional and non-functional requirements to derive test cases and verify whether the implemented system satisfies the specified requirements.

iv) End Users: final users can refer to the document to understand the major capabilities of the system, including complaint registration, categorization, tracking, notifications, and status monitoring.

v) Administrators: the admins who play an equally important role as the users, can make use of the document to understand the administrative functions of the system, including complaint assignment, escalation, status management, resolution, dashboard monitoring, and reporting.

vi) Project Team and Evaluators: the project team and evaluators can use this document to understand the system scope, requirements, analysis models, system features, and quality requirements.

Product Scope

*The project **Smart Complaint Management System** is a centralized platform for lodging, managing, tracking, and resolving complaints. The system provides users with a structured mechanism to register complaints and monitor their progress throughout the complaint lifecycle.*

The system enables users to:

- *Register and log in to the system.*
- *Lodge complaints.*
- *Categorize complaints.*
- *Track the status and progress of complaints.*
- *Receive notifications regarding complaint updates.*
- *Search and filter complaints.*

Whereas for the administrators , the system enables them to:

- *View and manage complaints.*
- *Assign complaints for resolution.*
- *Escalate complaints when required.*
- *Update complaint status.*
- *Resolve complaints.*
- *Monitor complaints through an administrative dashboard.*
- *Generate and view reports and analytics of the complaint resolutions made.*

  Aside from these, the system also provides smart complaint classification capabilities for priority categorization and duplicate complaint detection.

  The primary objective of the system is to improve the organization and efficiency of complaint management by providing a centralized platform for complaint handling, tracking, assignment, escalation, and resolution so it is hassle free and the process of making complaints is streamlined. It aims to improve transparency for users, assist administrators in managing complaints efficiently, and support timely identification and handling of high-priority or duplicate complaints.

References

[1] *Team 8, Smart Complaint Management System – Use Case Diagram, Version 1.0, 2026. (attached in github and also in the following sections)*
[2] *Jackfruit Phase-1 Software Requirements Specification Template, Version 1.0, provided by the Department of Software Engineering, August 2026.*
[3] *Team 8, Smart Complaint Management System – Requirement List (Functional and Non-Functional Requirements), Version 1.0, 2026. (attached in github)*
[4] *CPGRAMS (Centralized Public Grievance Redress and Monitoring System) for the UI inspiration*

Overall Description

Product Perspective

The Smart Complaint Management System is a self-contained web-based system being developed as part of this project. It is intended to provide a centralized platform for complaint registration, management, tracking, and resolution. The system is developed specifically according to the requirements defined for this project and is ***not intended as a replacement or extension of an existing organizational complaint management system****.*

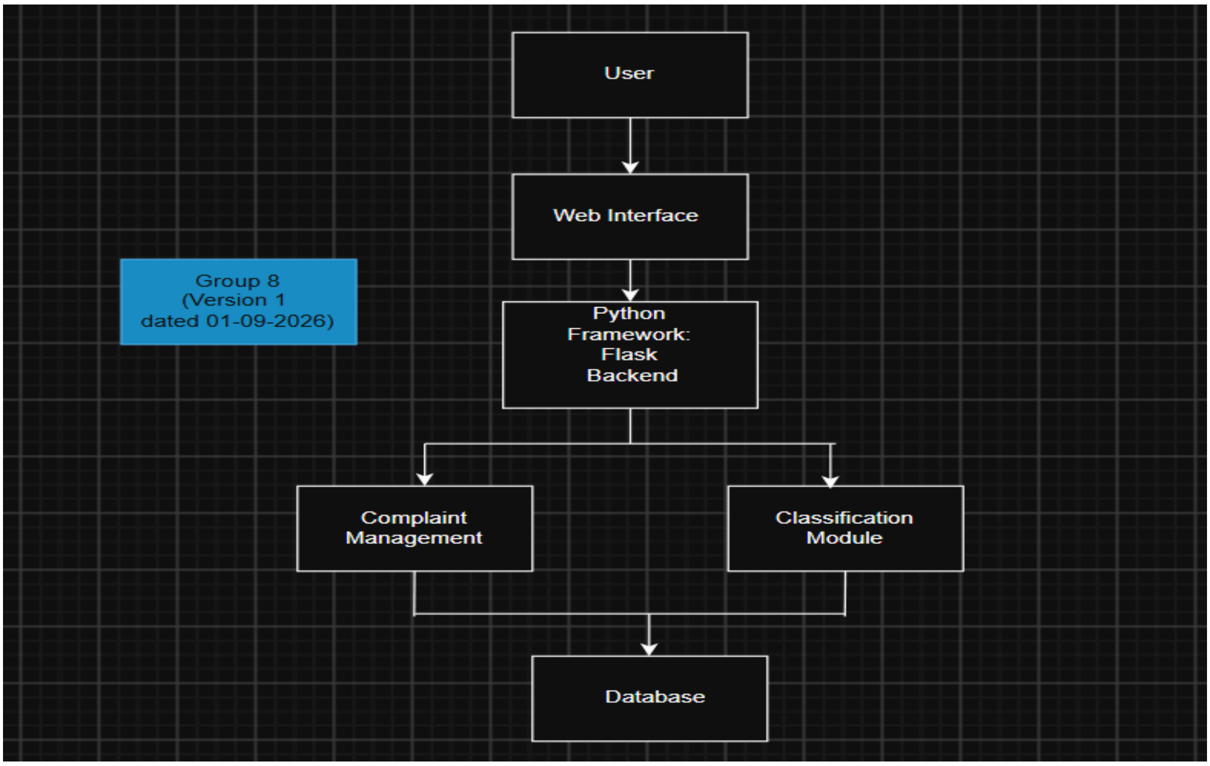
The system follows a client-server architecture in which users and administrators interact with the system through a web-based user interface. The application processes user requests, manages complaint information, applies complaint classification logic, maintains complaint status and assignment information, and provides appropriate responses and notifications.

The system consists of two primary user roles:

- ***User:*** *Registers or logs in to the system, lodges complaints, categorizes complaints, tracks complaints, receives notifications, and searches or filters complaints.*
- ***Administrator:*** *Manages complaints, assigns complaints, escalates complaints, updates complaint status, resolves complaints, and accesses administrative dashboards and reports.*

The system is intended to provide a centralized alternative to fragmented or manual complaint-handling processes by maintaining complaint information and its progress within a single platform.

A high-level architecture diagram may be included here to illustrate the interaction between users, the web application, the application logic, the database, and any external services used by the system.
The system will be implemented using Python as the primary programming language, with a suitable Python web framework Flask



Product Functions

The Smart Complaint Management System provides the following major functions:

User Management

- *User registration and login.*
- *Role-based access control.*

Complaint Management

- *Complaint registration.*
- *Complaint categorization.*
- *Complaint tracking.*
- *Complaint status updates.*
- *Complaint assignment.*
- *Complaint escalation.*
- *Complaint resolution.*

Notification Management

- *Notifications to users regarding relevant complaint updates and progress.*

Smart Complaint Classification

- *Priority categorization of complaints.*
- *Detection of duplicate complaints.*

Administrative Management

- *Administrative dashboard for monitoring complaint activities.*
- *Search and filtering of complaints.*
- *Reports and analytics for complaint management.*

These functions collectively support the complaint lifecycle from registration through categorization, assignment, tracking, escalation, status updates, and resolution.

User Classes and Characteristics

User Classes and Characteristics

The Smart Complaint Management System has two primary user classes: ***User*** *and* ***Administrator****.*

i) User

Users are individuals who access the system to submit and monitor complaints. They are expected to have basic computer and web browsing knowledge and do not require specialized technical expertise.

Users can:

- *Register and log in.*
- *Lodge complaints.*
- *Categorize complaints.*
- *Track complaints.*
- *Escalate complaints.*
- *Receive complaint-related notifications.*
- *Search and filter complaints.*
- *View complaint status and progress.*

ii) Administrator

Administrators are authorized personnel responsible for managing complaints and monitoring the complaint management process. Administrators require a higher level of system privileges than regular users.

Administrators can:

- *Access and manage complaints.*
- *Assign complaints.*
- *Update complaint status.*
- *Resolve complaints.*
- *Categorize and manage complaints.*
- *Access the administrative dashboard.*
- *View reports and analytics.*
- *Search and filter complaints.*

Role-based access control is used to ensure that users can access only the functions permitted for their respective roles.

Operating Environment

The Smart Complaint Management System is intended to operate as a web-based application accessible through commonly used web browsers.

Client Environment

- *Desktop or laptop computer capable of running a modern web browser.*
- *Modern web browser such as Google Chrome, Mozilla Firefox, Microsoft Edge, or equivalent.*
- *Internet/network connectivity for accessing the web application.*

Application Environment

The system will be implemented using one of the web development frameworks specified for the project:

- *Python with Flask*

Server Environment

The application will operate on a server capable of hosting the selected web framework and its required runtime environment.

Database Environment

***MySQL** will be used as the database management system for the Smart Complaint Management System. It will store and manage user information, complaint details, complaint categories, assignments, status updates, notifications, and other data required by the system*

Development Environment

Development and testing will be performed on standard computer systems capable of supporting Python, the Flask web framework, MySQL database management system, and the required web server components.

Design and Implementation Constraints

Technology Stack Constraint: *The system shall be implemented using* ***Python as the primary programming language, Flask as the web application framework, and MySQL as the database management system****.*

Time and Resource Constraint: *The project shall be developed and completed within the stipulated academic timeline and the available team resources.*

External Integration Constraint: *The system shall function as a standalone academic prototype and shall not depend on integration with external government, municipal, or third-party authentication systems. Administrative functions shall be handled through the system's administrator interface.*

Data Privacy Constraint: *User personal information, including name, contact details, and location information, shall be stored securely and shall not be unnecessarily exposed through publicly accessible complaint information.*

Scalability Constraint: *The system is designed as an academic proof-of-concept for a limited deployment environment and is not intended to support large-scale nationwide deployment as part of the current project.*

Browser Compatibility Constraint: *The web application shall support commonly used modern web browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge.*

Privacy and Security Constraint: *The system shall follow the defined privacy and security requirements by limiting the collection and exposure of personal information to what is necessary for complaint management and preventing access to such information by unauthorized users.*

2.6 Assumptions and Dependencies

Assumptions:

- *Users are assumed to have access to a computer or laptop with a supported modern web browser and a working internet connection.*
- *Users are assumed to provide valid information during registration and login and to maintain the confidentiality of their account credentials.*
- *Administrators are assumed to manually log in to the system to manage complaints, update complaint status, assign complaints, escalate complaints, and resolve complaints.*
- *The predefined complaint categories are assumed to cover the majority of complaints handled by the system. Complaints that do not fit an existing category may be classified under an appropriate general or "Other" category.*
- *Users are assumed to have basic digital literacy sufficient to navigate the web interface and submit and track complaints.*

Dependencies:

- *The system depends on the availability of the* ***Flask web framework and Python runtime environment*** *for application execution.*

- *The system depends on the availability and proper functioning of the **MySQL database** for storing and retrieving user, complaint, status, assignment, and related system data.*
- *The notification functionality depends on the availability of the configured notification mechanism used by the system for delivering complaint status and other relevant notifications to users.*
- *The system depends on a stable network connection for communication between users, the web application, and the database server.*

External Interface Requirements

User Interfaces

*The Smart Complaint Management System shall provide a **web-based user interface** that can be accessed through supported modern web browsers. Separate interface views shall be provided for users and administrators according to their authorized functions.*

User Interface:

- ***Registration and Login:** Provides users with forms for creating an account and securely logging into the system.*
- ***User Dashboard:** Provides access to complaint registration, previously submitted complaints, complaint tracking, and relevant notifications.*
- ***Complaint Registration Interface:** Provides a form through which users can enter the required complaint details and select an appropriate complaint category.*
- ***Complaint Tracking Interface:** Displays the current status and progress of submitted complaints to allow users to monitor their complaints.*
- ***Notification Interface:** Displays relevant notifications regarding complaint registration, assignment, status changes, escalation, and resolution.*

Administrator Interface:

- ***Administrator Login:** Provides secure access to administrative functions.*
- ***Administrator Dashboard:** Provides an overview of complaints and relevant complaint statistics.*
- ***Complaint Management Interface:** Allows administrators to view and manage complaints and access relevant complaint information.*
- ***Search and Filtering Interface:** Provides options to search and filter complaints based on available complaint attributes.*
- ***Complaint Assignment Interface:** Allows administrators to assign complaints to the appropriate personnel or department.*
- ***Complaint Status Management Interface:** Allows authorized administrators to update the status of complaints and manage their progress.*
- ***Complaint Escalation Interface:** Provides functionality for managing complaints that require escalation.*
- ***Reports and Analytics Interface:** Provides administrators with reports and analytical information related to complaints.*

General UI Guidelines:

- *The interface shall use a **simple, clear, and consistent layout** across all screens.*
- *Navigation elements shall be placed consistently to allow users to easily move between major functions such as the dashboard, complaint registration, complaint tracking, notifications, and logout.*
- *Standard buttons and controls such as **Submit, Cancel, Save, Back, Search, Filter, Update, and Logout** shall be used where applicable.*

- *Input fields shall provide appropriate labels and instructions to clearly indicate the information required from the user.*
- *The system shall perform appropriate **input validation** and display clear and understandable error messages when invalid, incomplete, or incorrect information is entered.*
- *Success and confirmation messages shall be displayed after important operations such as successful registration, complaint submission, status updates, and other relevant actions.*
- *Destructive or important actions shall require appropriate confirmation before they are completed where necessary.*
- *The interface shall provide clear visual indication of the **current complaint status** so that users can understand the progress of their complaints.*
- *Text, buttons, forms, and navigation elements shall be presented in a readable and consistent manner to support ease of use.*
- *The interface shall provide appropriate accessibility considerations, including **readable text, clear labels, sufficient contrast, and keyboard-accessible controls**.*
- *The interface shall be designed for use on **supported desktop and laptop web browsers** and shall maintain a consistent layout across commonly used modern browsers.*
- *Access to screens and functions shall be controlled according to the user's **authorized role**.*

*The user interface may take inspiration from the general layout and navigation conventions of existing **CPGRAMS (Centralized Public Grievance Redress and Monitoring System)** . The referenced interfaces are used only as design references; the final interface shall be developed according to the requirements of the Smart Complaint Management System*

Software Interfaces

The Smart Complaint Management System shall interact with the following software components:

Operating Environment:
The system shall operate in a supported server/development environment capable of running the Python runtime, Flask framework, and MySQL database. Users shall access the system through supported modern web browsers.

Backend Framework:
Flask *shall be used as the web application framework for implementing the server-side logic of the system. It shall handle requests from the web interface, process application logic, communicate with the database, and return the required responses to the user interface.*

Database:
MySQL *shall be used for persistent storage and retrieval of system data. The database shall store information including user details, complaint details, complaint categories, complaint status, assignments, escalation information, and other data required for complaint management.*

Web User Interface:
The system shall provide a browser-based interface through which users and administrators interact with the application. The interface shall communicate with the Flask backend to submit and retrieve user and complaint-related information.

Authentication and Authorization:
The system shall provide authentication functionality for user registration and login. Role-based access control shall be used to restrict access to administrative functions and other protected system features according to the user's authorized role.

Notification Interface:
The system shall use the configured notification mechanism to deliver relevant notifications to users regarding complaint submission, status changes, escalation, resolution, and other applicable complaint events.

Data Exchange:
Data exchanged between the web interface, Flask backend, and MySQL database shall include user information, complaint details, complaint categories, complaint status, assignment information, escalation information, and notification-related information. The Flask backend shall process requests and responses between the user interface and the database.

Communications Interfaces

The Smart Complaint Management System shall use standard web communication protocols to facilitate communication between users, administrators, the web browser, the Flask application, and the database server.

Client–Server Communication:
Communication between the user's web browser and the Flask-based web application shall use ***HTTP/HTTPS****, with* ***HTTPS preferred for deployment*** *to protect data transmitted between the client and server.*

Web Communication:
The system shall support standard web communication between the client browser and the application server for operations such as user registration, login, complaint submission, complaint tracking, complaint management, status updates, search, filtering, and administrative operations.

Data Transfer:
User and complaint information shall be transmitted between the web interface and the application

server using standard web request and response mechanisms. The data exchanged shall include user details, complaint information, category information, status information, assignment details, and other information required for system operations.

Notification Communication:
The system shall provide notifications to users regarding relevant complaint events, including complaint submission, status changes, escalation, and resolution. The specific notification delivery mechanism shall depend on the notification service configured during implementation.

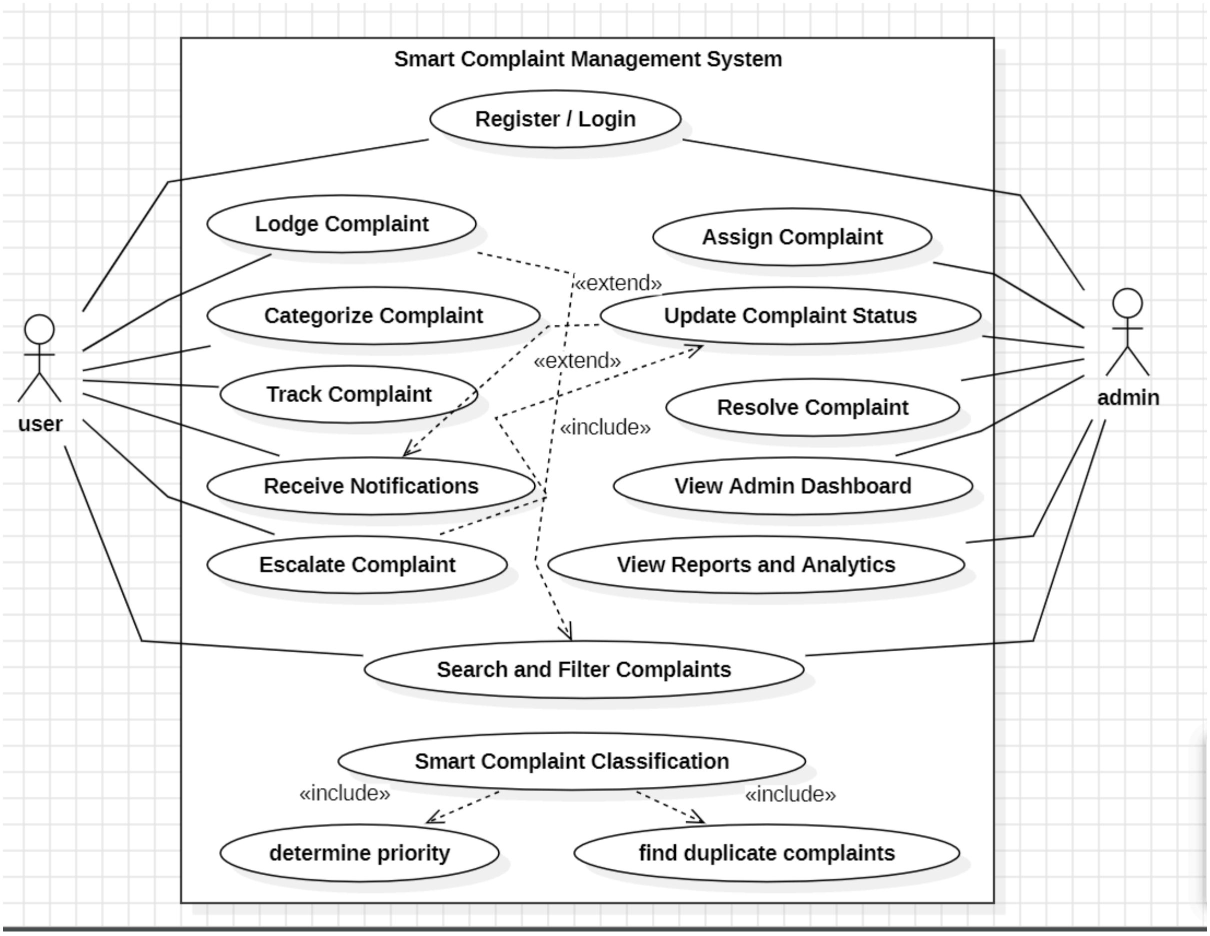
Network Requirement:
A stable internet connection shall be required to access the web application and perform system operations. The system shall function through commonly used modern web browsers.

Communication Security:
*Communication containing user credentials and complaint-related information shall be protected using secure communication mechanisms such as **HTTPS** during deployment. The system shall prevent unauthorized access to transmitted information.*

(WE WILL NOT BE IMPLEMENTING HARDWARE INTERFACE FOR THIS PROJECT)

Analysis Models

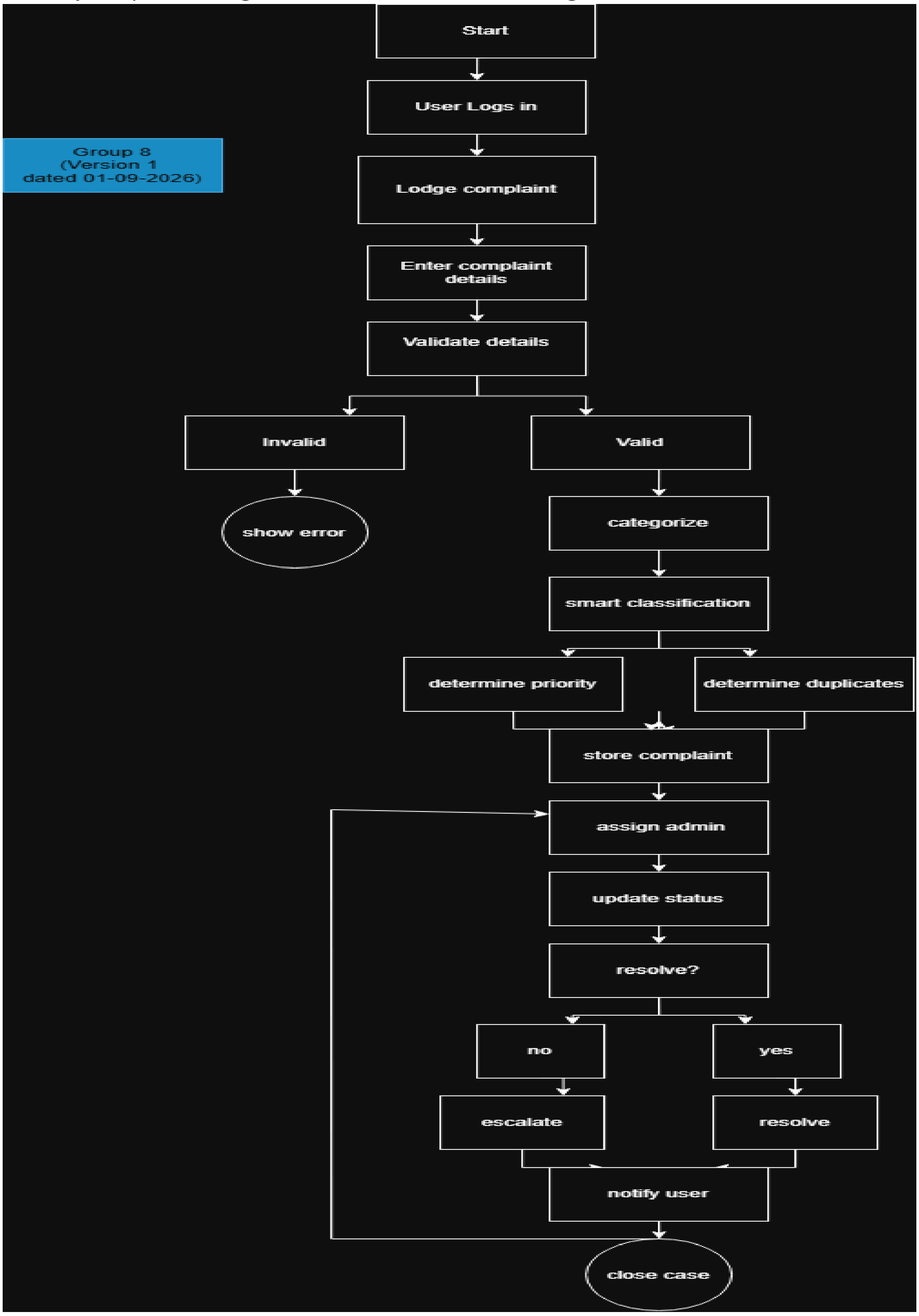
Use Case Diagram:



The use case diagram illustrates the interactions between the two primary actors, User and Administrator, and the major functions provided by the Smart Complaint Management System. Users primarily interact with complaint registration, categorization, tracking, notifications, and search functionality, while administrators manage assignment, escalation, status updates, resolution, dashboards, and reports. Smart complaint classification supports priority determination and duplicate complaint detection

Activity Diagram:

The activity diagram represents the workflow of complaint processing in the Smart Complaint Management System. It illustrates the sequence of activities beginning with user authentication and complaint registration, followed by validation, categorization, smart classification, complaint management, assignment, status updates, escalation when required, resolution, and notification of the *user finally ending with the case being marked as resolved and closed.*



System Features

5.1 User Registration and Login

5.1.1 Description and Priority

This feature allows users to create an account and securely log in to the Smart Complaint Management System. Authentication is required to access protected system functions such as complaint registration, complaint tracking, and other authorized user or administrative functions.

Priority: High Benefit: 9, Penalty: 9, Cost: 5, Risk: 3

5.1.2 Stimulus/Response Sequences

- *User selects **"Register"** → System displays the registration form.*
- *User enters the required registration details and submits the form → System validates the provided information.*
- *Registration information is valid → System creates the user account and displays a registration confirmation.*
- *User enters valid login credentials and selects **"Login"** → System authenticates the user and provides access to the appropriate dashboard.*
- *User enters invalid login credentials → System rejects the login attempt and displays an appropriate error message.*
- *An unauthenticated user attempts to access a protected function → System denies access and prompts the user to log in.*

5.1.3 Functional Requirements

REQ-1.1: *The system shall allow a new user to register by providing the required registration information.*

REQ-1.2: *The system shall validate the registration information before creating a user account.*

REQ-1.3: *The system shall prevent registration using information that is already associated with an existing account and shall display an appropriate error message.*

REQ-1.4: *The system shall allow a registered user to log in using valid authentication credentials.*

REQ-1.5: *The system shall deny access when invalid login credentials are provided and shall display an appropriate error message.*

REQ-1.6: *The system shall restrict access to protected system functions to authenticated users.*

REQ-1.7: *The system shall provide access to system functions based on the authenticated user's authorized role.*

5.2 Complaint Registration

5.2.1 Description and Priority

This feature allows a user to register a new complaint about an issue by providing the required complaint details and selecting an appropriate complaint category. It is a core data-entry feature of the system and provides the information required for subsequent complaint categorization, classification, assignment, tracking, and resolution.

Priority: High *Benefit: 9, Penalty: 9, Cost: 5, Risk: 3*

5.2.2 Stimulus/Response Sequences

- *User selects "Lodge Complaint" → System displays the complaint registration form.*
- *User enters the required complaint details and selects an appropriate category → System accepts the entered information.*
- *User submits the complaint form → System validates all required information.*
- *Complaint information is valid → System stores the complaint and displays a confirmation message.*
- *User submits an incomplete or invalid form → System highlights the relevant fields and displays an appropriate error message.*

5.2.3 Functional Requirements

REQ-2.1: The system shall provide a complaint registration form through which users can enter the required complaint details.

REQ-2.2: The system shall allow users to select an appropriate category for the complaint during complaint registration.

REQ-2.3: The system shall validate the required complaint information before accepting the complaint.

REQ-2.4: The system shall reject complaint registration when required information is missing or invalid and shall display an appropriate error message.

REQ-2.5: The system shall store successfully submitted complaint information in the database.

REQ-2.6: The system shall display a confirmation message to the user after successful complaint registration.

5.3 Complaint Categorization

5.3.1 Description and Priority

This feature allows complaints to be classified into predefined complaint categories so that complaints can be organized and managed appropriately. The assigned category helps in the subsequent processing, assignment, tracking, and resolution of complaints.

Priority: Medium *Benefit: 7, Penalty: 5, Cost: 5, Risk: 4*

5.3.2 Stimulus/Response Sequences

- *User enters the required complaint details → System displays the available complaint categories.*
- *User selects an appropriate complaint category → System associates the selected category with the complaint.*
- *User submits the complaint → System validates and stores the complaint along with its category.*
- *A complaint requires categorization or correction → System allows an authorized administrator to assign or update the appropriate category.*

- *An invalid or unavailable category is selected → System displays an appropriate error message and prompts the user or administrator to select a valid category.*

5.3.3 Functional Requirements

REQ-3.1: The system shall provide a predefined list of complaint categories for classifying complaints.

REQ-3.2: The system shall allow users to select an appropriate complaint category while registering a complaint.

REQ-3.3: The system shall associate the selected category with the corresponding complaint and store it in the database.

REQ-3.4: The system shall allow an authorized administrator to update the category of a complaint when required.

REQ-3.5: The system shall validate the selected complaint category before storing or updating the complaint.

5.4 Complaint Tracking and Notifications

5.4.1 Description and Priority

This feature allows users to monitor the progress of their submitted complaints and receive relevant notifications regarding changes to their complaints. It enables users to view the current status and track the progress of their complaints throughout the complaint management process.

***Priority: High** Benefit: 8, Penalty: 6, Cost: 4, Risk: 2*

5.4.2 Stimulus/Response Sequences

- *User opens "My Complaints" → System displays a list of complaints submitted by the user along with their current status.*
- *User selects a specific complaint → System displays the complaint details and its current status.*
- *Administrator updates the status of a complaint → System updates the complaint information and generates a relevant notification for the user.*
- *User opens a notification → System displays the related complaint information and its updated status.*

5.4.3 Functional Requirements

REQ-4.1: The system shall allow users to view the current status of their submitted complaints.

REQ-4.2: The system shall maintain complaint status information throughout the complaint management process.

REQ-4.3: The system shall provide relevant notifications to users when significant events occur in relation to their complaints, including status updates, escalation, and resolution.

REQ-4.4: The system shall allow users to view their previously submitted and active complaints through the complaint tracking interface.

REQ-4.5: The system shall display relevant complaint information, including complaint details, category, and current status, when a user selects a complaint for tracking.

5.5 Complaint Assignment

5.5.1 Description and Priority

This feature allows authorized administrators to assign complaints to the appropriate personnel or department responsible for handling them. It ensures that submitted complaints are directed to the appropriate authority for further processing and resolution.

Priority: High *Benefit: 8, Penalty: 7, Cost: 5, Risk: 4*

5.5.2 Stimulus/Response Sequences

- *Administrator views the list of submitted complaints → System displays the available complaints for assignment.*
- *Administrator selects a complaint → System displays the relevant complaint details and available assignment options.*
- *Administrator selects the appropriate personnel or department → System associates the complaint with the selected assignee.*
- *Administrator confirms the assignment → System stores the assignment information and updates the complaint record.*
- *Administrator changes an existing assignment → System updates the complaint with the new assignment information.*

5.5.3 Functional Requirements

REQ-5.1: The system shall allow an authorized administrator to assign a complaint to the appropriate personnel or department.

REQ-5.2: The system shall display the relevant complaint information required for making an assignment.

REQ-5.3: The system shall store the assignment information associated with each complaint.

REQ-5.4: The system shall allow an authorized administrator to update or reassign a complaint when required.

REQ-5.5: The system shall update the complaint record whenever an assignment is created or modified.

5.6 Complaint Escalation

5.6.1 Description and Priority

This feature allows a user to escalate a complaint when they feel that the complaint has not received sufficient attention or appropriate action. The escalation brings the complaint to the attention of the appropriate administrator for further review and action.

Priority: Medium *Benefit: 7, Penalty: 6, Cost: 4, Risk: 3*

5.6.2 Stimulus/Response Sequences

- *User views a submitted complaint → System displays the current complaint status and available actions.*
- *User feels that the complaint has not received sufficient attention or appropriate action → User selects the "Escalate Complaint" option.*
- *System records the escalation request and updates the complaint's escalation status.*
- *Administrator views the escalated complaint → System displays the complaint details and escalation information.*
- *Administrator reviews the escalated complaint and takes appropriate action → System updates the complaint information accordingly.*

5.6.3 Functional Requirements

REQ-6.1: The system shall allow a user to escalate a submitted complaint when the user feels that the complaint has not received sufficient attention or appropriate action.

REQ-6.2: The system shall record the escalation request and associate it with the corresponding complaint.

REQ-6.3: The system shall update the complaint to indicate that it has been escalated.

REQ-6.4: The system shall allow authorized administrators to view and review escalated complaints.

REQ-6.5: The system shall allow administrators to take further action on escalated complaints and update their status accordingly.

5.7 Smart Complaint Classification (Priority Categorization)

5.7.1 Description and Priority

This feature automatically determines the priority of a complaint based on the information provided in the complaint. It helps identify complaints that require greater attention and supports administrators in managing complaints according to their urgency.

***Priority: Medium** Benefit: 8, Penalty: 5, Cost: 6, Risk: 5*

5.7.2 Stimulus/Response Sequences

- *A complaint is submitted → System analyzes the available complaint information to determine its priority.*
- *System determines the priority level → System assigns the appropriate priority to the complaint.*
- *Administrator views the complaint → System displays the assigned priority along with the complaint information.*
- *Administrator determines that the assigned priority is inappropriate → Administrator updates the priority level → System stores the updated priority.*

5.7.3 Functional Requirements

REQ-7.1: The system shall automatically determine and assign a priority level to each complaint based on the information available for the complaint.

REQ-7.2: The system shall classify complaints according to predefined priority levels.

REQ-7.3: The system shall associate the determined priority level with the corresponding complaint and store it in the database.

REQ-7.4: The system shall allow an authorized administrator to manually update the priority level of a complaint when required.

REQ-7.5: The system shall display the assigned priority level along with the relevant complaint information to authorized administrators

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5.8 Duplicate Complaint Detection

5.8.1 Description and Priority

This feature identifies complaints that may refer to the same underlying issue by comparing relevant complaint information. It helps reduce redundant complaint processing and allows administrators to identify complaints that may be related to an existing complaint.

Priority: Medium *Benefit: 6, Penalty: 4, Cost: 6, Risk: 6*

5.8.2 Stimulus/Response Sequences

- *A new complaint is submitted → System compares the complaint information with existing complaints to identify possible duplicates.*
- *A possible duplicate is detected → System identifies the related complaint and provides the information to the system for further handling.*
- *Administrator views the complaint → System displays information about possible duplicate complaints, if detected.*
- *Administrator reviews the detected duplicate → Administrator determines whether the complaints are related and takes the appropriate action.*

5.8.3 Functional Requirements

REQ-8.1: The system shall analyze relevant information from a newly submitted complaint and compare it with existing complaints to identify possible duplicates.

REQ-8.2: The system shall identify and display complaints that are likely to be duplicates of the newly submitted complaint.

REQ-8.3: The system shall retain each complaint as an individual user submission while identifying possible relationships between duplicate complaints.

REQ-8.4: The system shall provide administrators with information about possible duplicate complaints for review.

REQ-8.5: The system shall allow an authorized administrator to review and take appropriate action on complaints identified as possible duplicates.

5.9 Complaint Status Updates

5.9.1 Description and Priority

This feature allows authorized administrators to update the status of complaints as they progress through the complaint management process. It ensures that the current state of each complaint is maintained and allows users to track the progress of their submitted complaints.

Priority: High *Benefit: 8, Penalty: 6, Cost: 3, Risk: 2*

5.9.2 Stimulus/Response Sequences

- *Administrator views a complaint → System displays the current status and relevant complaint information.*
- *Administrator updates the complaint status → System validates the status update and stores the updated status.*
- *Status update is successfully recorded → System reflects the updated status in the complaint tracking interface.*
- *User views a submitted complaint → System displays its current status.*
- *An unauthorized user attempts to update a complaint status → System denies the action and displays an appropriate access error.*

5.9.3 Functional Requirements

REQ-9.1: The system shall maintain the current status of each complaint throughout the complaint management process.

REQ-9.2: The system shall allow only authorized administrators to update the status of a complaint.

REQ-9.3: The system shall provide predefined complaint status values for managing the progress of complaints.

REQ-9.4: The system shall store the updated status information associated with the corresponding complaint.

REQ-9.5: The system shall display the current complaint status to the user through the complaint tracking interface.

REQ-9.6: The system shall prevent unauthorized users from modifying complaint status information.

5.10 Admin Dashboard

5.10.1 Description and Priority

This feature provides administrators with a central interface to view, manage, and monitor complaints within the system. The dashboard provides access to relevant complaint information and administrative functions such as complaint assignment, status updates, escalation management, and search and filtering.

Priority: High *Benefit: 8, Penalty: 6, Cost: 5, Risk: 3*

5.10.2 Stimulus/Response Sequences

- *Administrator logs in → System authenticates the administrator and displays the administrator dashboard.*
- *Administrator opens the dashboard → System displays relevant complaint information and available administrative functions.*
- *Administrator selects a complaint → System displays the complaint details and available management actions.*
- *Administrator performs an administrative action such as assigning a complaint, updating its status, or managing an escalation → System validates and records the action.*
- *An unauthorized user attempts to access the administrator dashboard → System denies access to the dashboard and displays an appropriate access error.*

5.10.3 Functional Requirements

REQ-10.1: The system shall provide an administrator dashboard for managing and monitoring complaints.

REQ-10.2: The system shall allow authorized administrators to view relevant complaint information from the dashboard.

REQ-10.3: The system shall allow authorized administrators to access complaint management functions, including complaint assignment, status updates, and escalation management.

REQ-10.4: The system shall provide access to search and filtering functions through the administrator dashboard.

REQ-10.5: The system shall restrict access to the administrator dashboard to authenticated and authorized administrator accounts.

REQ-10.6: The system shall prevent unauthorized users from accessing administrator dashboard functions

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5.11 Report and Analytics

5.11.1 Description and Priority

This feature allows administrators to generate and view reports and analytical information based on complaint data. It helps administrators understand complaint patterns, monitor complaint status, and support informed decision-making and management of complaints.

Priority: Low *Benefit: 6, Penalty: 3, Cost: 6, Risk: 4*

5.11.2 Stimulus/Response Sequences

- *Administrator opens the "Reports and Analytics" section → System retrieves the available complaint data and displays relevant reports and analytical information.*
- *Administrator selects a report or applies an available filter → System processes the relevant complaint data and displays the corresponding results.*
- *Administrator views the generated report → System presents the complaint information in an understandable format.*
- *Administrator requests updated analytical information → System processes the latest available complaint data and displays the updated results.*

5.11.3 Functional Requirements

REQ-11.1: The system shall allow authorized administrators to generate reports based on available complaint data.

REQ-11.2: The system shall provide analytical information related to complaints, including complaint counts and complaint status information.

REQ-11.3: The system shall allow administrators to view complaint data based on available categories and status information.

REQ-11.4: The system shall present generated reports and analytical information in a clear and understandable format.

REQ-11.5: The system shall use the available complaint data to provide updated reports and analytical information when requested by an authorized administrator.

5.12 Role-Based Access Control

5.12.1 Description and Priority

This feature ensures that users and administrators can access only the system functions and information authorized for their respective roles. It protects restricted administrative functions and ensures that users can access only the complaint-related functions permitted to them.

Priority: High *Benefit: 9, Penalty: 8, Cost: 4, Risk: 5*

5.12.2 Stimulus/Response Sequences

- *A user or administrator logs in → System authenticates the account and identifies the user's authorized role.*
- *User accesses the system → System provides access only to functions permitted for the user role.*
- *Administrator accesses the system → System provides access to authorized administrative functions such as complaint management, assignment, status updates, escalation management, and reports and analytics.*
- *A user attempts to access an administrator-only function → System denies access and displays an appropriate authorization error.*
- *An unauthenticated user attempts to access a protected function → System denies access and prompts the user to log in.*

5.12.3 Functional Requirements

REQ-12.1: The system shall support role-based access control for users and administrators.

REQ-12.2: The system shall restrict users to functions and complaint information authorized for their role.

REQ-12.3: The system shall provide administrators with access to authorized administrative functions.

REQ-12.4: The system shall prevent users from accessing administrator-only functions.

REQ-12.5: The system shall enforce authorization checks before allowing access to protected system functions and data.

REQ-12.6: The system shall deny access to protected functions when the authenticated user does not have the required authorization.

5.13 Search and Filtering

5.13.1 Description and Priority

This feature allows users and administrators to efficiently locate relevant complaints using available search and filtering options. It helps users find their complaints and enables administrators to locate and manage complaints more efficiently.

Priority: Low *Benefit: 5, Penalty: 3, Cost: 3, Risk: 2*

5.13.2 Stimulus/Response Sequences

- *User or administrator opens the search and filtering interface → System displays the available search and filtering options.*
- *User or administrator enters a search term or selects a filter → System processes the provided criteria.*
- *Matching complaints are found → System displays the complaints that satisfy the specified search or filtering criteria.*
- *Search or filtering criteria return no matching complaints → System displays a clear "No results found" message.*
- *User or administrator clears the search or filtering criteria → System displays the available complaints according to the user's authorized access.*

5.13.3 Functional Requirements

REQ-13.1: The system shall provide search functionality for locating relevant complaints.

REQ-13.2: The system shall allow users to search and filter complaints that they are authorized to access.

REQ-13.3: The system shall allow administrators to search and filter complaints available to them based on relevant complaint attributes.

REQ-13.4: The system shall support filtering of complaints using available complaint attributes such as category and status.

REQ-13.5: The system shall display the complaints that match the specified search or filtering criteria.

REQ-13.6: The system shall display a clear "No results found" message when no complaints match the specified search or filtering criteria.

Nonfunctional Requirements Relevant to the Smart Features (7–13)

- ***Accuracy:*** *The smart complaint classification and duplicate complaint detection features shall provide sufficiently accurate results during testing to support appropriate complaint prioritization and identification of potentially duplicate complaints.*

- ***Performance:*** *The search and filtering functionality shall provide results within a reasonable response time under the typical dataset size used during system testing and demonstration.*
- ***Security:*** *Role-based access control shall be enforced at the application level to ensure that users can access only the functions and complaint information authorized for their respective roles.*
- ***Auditability:*** *The system shall maintain relevant records of important complaint management activities, including complaint status updates, assignments, and escalations, to support accountability and traceability.*
- ***Usability:*** *The administrator dashboard and reports and analytics interfaces shall present complaint information in a clear, organized, and understandable manner to support efficient complaint management.*
- ***Data Integrity:*** *The system shall maintain the accuracy and consistency of complaint information when complaints are categorized, classified, assigned, updated, escalated, or processed as possible duplicates.*

Other Nonfunctional Requirements

1. Performance Requirements

The system shall provide responsive performance during normal operation so that users and administrators can efficiently perform complaint management activities.

- ***NFR-P1:*** *The system shall process complaint registration requests within a reasonable response time under the typical dataset size used during testing and demonstration.*
- ***NFR-P2:*** *The system shall provide complaint tracking information without unnecessary delay after the user requests it.*
- ***NFR-P3:*** *The administrator dashboard shall load complaint information and relevant statistics within a reasonable response time under the expected prototype workload.*
- ***NFR-P4:*** *Search and filtering operations shall return matching complaint results within a reasonable response time under the expected prototype workload.*
- ***NFR-P5:*** *The system shall process complaint status updates and reflect the updated status in the relevant interfaces without unnecessary delay.*

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2. Safety Requirements

The Smart Complaint Management System is a web-based information and complaint-management system and does not directly control physical equipment or safety-critical processes.

- ***NFR-S1:*** *The system shall prevent accidental loss or modification of complaint information by validating important operations before they are processed.*
- ***NFR-S2:*** *The system shall preserve complaint information required for complaint tracking, management, and resolution.*
- ***NFR-S3:*** *The system shall display appropriate error messages when invalid or incomplete information is entered so that users can correct the information before submission.*
- ***NFR-S4:*** *The system shall provide appropriate confirmation for important complaint-management operations where accidental actions could affect complaint information.*
- ***NFR-S5:*** *No formal safety certification is required for the current academic prototype because the system does not operate or control safety-critical physical systems.*

3. Security Requirements

- ***NFR-SEC1:*** *The system shall require users to authenticate using valid account credentials before accessing protected system functions.*
- ***NFR-SEC2:*** *The system shall protect user authentication credentials and shall not store passwords in plain text.*
- ***NFR-SEC3:*** *The system shall use secure communication mechanisms such as HTTPS during deployment to protect user credentials and complaint-related information transmitted between the client and server.*
- ***NFR-SEC4:*** *The system shall enforce role-based access control at the application level so that users cannot access administrator-only functions or unauthorized complaint information.*
- ***NFR-SEC5:*** *The system shall restrict access to personal and complaint-related information to authorized users and administrators according to their roles.*
- ***NFR-SEC6:*** *The system shall prevent unauthenticated users from accessing protected system functions.*
- ***NFR-SEC7:*** *The system shall maintain records of relevant administrative activities, including complaint assignment, status updates, and escalation-related actions, to support accountability.*
- ***NFR-SEC8:*** *The system shall follow appropriate security and privacy practices for protecting user and complaint information within the scope of the academic prototype.*

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4. Software Quality Attributes

- ***Usability:*** *The system shall provide a simple, clear, and consistent web interface that allows users to register, lodge complaints, track complaints, and view notifications without requiring specialized training.*
- ***Availability:*** *The system shall be available and functional during scheduled testing, demonstration, and evaluation periods.*
- ***Reliability:*** *The system shall handle invalid inputs and unexpected application conditions gracefully without crashing or causing loss of valid complaint information.*
- ***Scalability:*** *The system shall use a design that allows the application and database to be extended to support an increased number of users and complaints beyond the current prototype scale.*
- ***Maintainability:*** *The system shall use a structured codebase with clearly organized application, database, and interface components so that individual features can be modified or maintained without unnecessarily affecting unrelated functionality.*
- ***Data Integrity:*** *The system shall maintain accurate and consistent user, complaint, category, assignment, status, escalation, and notification information throughout the complaint management process.*
- ***Accuracy:*** *The system shall provide sufficiently accurate results for smart complaint classification and duplicate complaint detection during system testing.*
- ***Fault Tolerance:*** *The system shall handle application and input errors gracefully and shall provide appropriate error messages without terminating the entire application.*
- ***Compatibility:*** *The web application shall function correctly on commonly used modern web browsers, including Google Chrome, Mozilla Firefox, and Microsoft Edge.*
- ***Auditability:*** *The system shall maintain relevant records of important complaint-management activities, including assignments, status updates, and escalations, to support traceability and accountability.*

- ***Privacy:*** *The system shall limit access to personal information and complaint-related data to authorized users and administrators and shall avoid unnecessary exposure of such information.*
- ***Accessibility:*** *The system shall provide readable text, clear labels, understandable navigation, and appropriately designed interface controls to support users with different levels of digital familiarity.*
- ***Relative Priority:*** *For this system, ease of use and reliability are prioritized over advanced customization, as users should be able to lodge and track complaints through a straightforward and understandable interface.*

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5. Business Rules

- ***BR1:*** *Only registered and authenticated users shall be permitted to lodge complaints through the system.*
- ***BR2:*** *Users shall be permitted to view and track only complaints that they are authorized to access.*
- ***BR3:*** *Only authorized administrators shall be permitted to assign complaints to personnel or departments.*
- ***BR4:*** *Only authorized administrators shall be permitted to update the status of complaints.*
- ***BR5:*** *A user shall be permitted to escalate their own submitted complaint when they feel that the complaint has not received sufficient attention or appropriate action.*
- ***BR6:*** *Escalated complaints shall be made available to authorized administrators for review and further action.*
- ***BR7:*** *Only authorized administrators shall be permitted to access administrative functions such as complaint management, assignment, status updates, escalation management, and reports and analytics.*
- ***BR8:*** *The system shall maintain each complaint as an individual complaint record and shall preserve its relevant information throughout the complaint management process.*
- ***BR9:*** *Complaint information and user information shall be accessible only according to the user's authorized role.*

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6. Domain Requirements

The following requirements are derived from the complaint-management domain and define general principles that the system shall follow when handling civic complaints.

- ***DR1:*** *Complaints shall be organized using predefined categories relevant to the types of issues handled by the complaint management system.*
- ***DR2:*** *Each complaint shall maintain information about its current status so that its progress can be tracked throughout the complaint management process.*
- ***DR3:*** *Complaints shall be assignable to the appropriate personnel or department for further processing and resolution.*
- ***DR4:*** *A complaint that requires additional attention shall be capable of being escalated by the user who submitted it.*
- ***DR5:*** *The system shall support identification of potentially duplicate complaints so that related complaints can be recognized during complaint management.*
- ***DR6:*** *Complaint information shall be maintained consistently throughout categorization, classification, assignment, status updates, escalation, and resolution.*

- ***DR7:*** *Personal information associated with complaints shall be protected and shall not be unnecessarily exposed to unauthorized users.*

7. Other Requirements

- ***Legal/Compliance Requirement:*** *The system shall clearly state that it is an academic prototype developed as part of the project and is not an official government or municipal complaint management platform.*
- ***Scope Exclusion Requirement:*** *The system shall clearly indicate that emergency situations and serious crimes requiring immediate intervention are outside the scope of the system and should be reported to the appropriate emergency or law-enforcement services.*
- ***Data Retention Requirement:*** *The system shall retain complaint records and related information for as long as required for complaint tracking, resolution, reporting, and system evaluation. The specific retention period may be defined according to the deployment requirements.*
- ***Database Requirement:*** *The system shall use a MySQL database to store and manage user information, complaint details, categories, assignments, status information, escalation records, notification-related information, and other data required for system operation.*
- ***Data Backup Requirement:*** *The system should support appropriate backup and recovery procedures for preserving complaint and user data and reducing the risk of data loss.*
- ***Reuse Requirement:*** *Common software components such as forms, navigation elements, status indicators, and notification components should be designed for reuse wherever applicable to improve consistency and reduce development effort.*
- ***Future Enhancement Requirement:*** *Possible future enhancements may include integration with external municipal or organizational complaint management systems, multilingual support, voice-based complaint filing, chatbot-based complaint assistance, and other advanced communication channels. These enhancements are outside the scope of the current project.*

Appendix A: Glossary

Term	Definition
Complaint	A formal issue or grievance submitted by a registered user through the system.
User	A registered person who can lodge, track, and manage their own complaints through the system.
Administrator	An authorized user responsible for managing complaints and performing administrative functions.
Complaint Category	A predefined classification used to organize complaints according to their type or nature.
Complaint Status	The current state of a complaint during its processing and resolution.
Complaint Assignment	The process of assigning a complaint to the appropriate personnel or department for handling.

Complaint Escalation	A user-initiated action used when the user feels that their complaint has not received sufficient attention or appropriate action.
Priority	The level of importance or urgency assigned to a complaint by the smart complaint classification feature.
Smart Complaint Classification	A system feature that automatically determines the priority of a complaint based on the available complaint information.
Duplicate Complaint	A complaint that may refer to the same underlying issue as another complaint.
Duplicate Complaint Detection	The process of identifying complaints that may refer to the same underlying issue by comparing relevant complaint information.
Role-Based Access Control (RBAC)	A security mechanism that restricts system functions and information according to the user's authorized role.
Notification	Information provided to a user regarding relevant complaint events such as submission, status changes, escalation, or resolution.
Admin Dashboard	A centralized interface through which authorized administrators can view and manage complaints and related administrative functions.
Reports and Analytics	Information generated from complaint data to help administrators understand complaint patterns and monitor complaint-related information.
Search and Filtering	Functionality that allows users and administrators to locate relevant complaints using available search and filtering criteria.
Flask	The Python-based web application framework used to develop the backend of the system.
MySQL	The relational database management system used to store and manage system data.

Appendix B: Field Layouts

An Excel sheet containing field layouts and properties/attributes and report requirements.

Sample sheet with information required to register the customer

Field	**Length**	**Data Type**	**Description**	**Is Mandatory**

Account Number	16	Numeric		Y
ISFC code	11	Alphanumeric		Y
Card Amount	20	Numeric		Y
Mandate Start Date	8	Date	Date of Mandate Registration	N
Mandate End Date	8	Date	Date of Mandate Expiry	N
Status	25	Alphanumeric	Status of Registration	Y
Customer Name	60	String		Y
Reject Reason Code	4	String	Reject Reason code in case mandate is rejected	N

Sample Report Requirements: Include the fields to be included in the report

Registration Report	**Transaction Report**
Bank Account Number	Transaction Reference Number
ISFC Code	Bank Account Number
Bank Name	IFSC Code
Account Status	Bank Name
Account Type	Customer Name
Customer Name	Card Number
Card Number	Debit Transaction Amount
SI Start Date	Transaction Date
Status	Status
Remarks	Debit Attempt Number
	Remarks

Appendix C: Requirement Traceability Matrix

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID

UT-01	User registration module	To test the login functionality	Access to Chrome Browser	1: Navigate to http://www.demo.com 2: Enter Username and Password 3: Click Submit	User name: PESU Student. Password : pes123	Login should be successful with “welcome message”	Login successful with “welcome message” displayed	Pass